

NAME

Solomon Shekpami Kyauta

STUDENT ID

20253812

LAB IV

General Objectives:

The Software Engineering Lab has been developed by keeping in mind the following objectives:

- To impart state-of-the-art knowledge on Software Engineering and UML in an interactive manner through the Web.
- Present case studies to demonstrate practical applications of different concepts.
- Provide a scope to students where they can solve small, real-life problems.

Lab Objectives:

By the end of this lab, students will be able to:

- Translate your class diagram from the previous lab into working Java code.
- Understand how each part of your UML design connects to your implementation.
- Apply object-oriented principles (encapsulation, inheritance, abstraction, and polymorphism).
- Strengthen your understanding of how software design becomes a functioning system.

Precautions:

- Use previous knowledge from SWE202 and other online sources except an AI chatbot.
- **Do not copy your friend's work, and avoid direct AI insertion.**
- Use actual collaborative tools/frameworks/IDEs where necessary. You may use Java, Python, or any other object-oriented language, but your code should follow object-oriented principles as shown in your UML diagram
- Use this exact manual, At the end of the day, convert it into a **PDF** for the final submission.
- Answer each question on a **fresh page**.

The Scenario (Note that this scenario may be used for the entire Lab:

Near East University Hospital wants to implement a Remote Health Monitoring System to track patients with chronic conditions such as hypertension and diabetes. Patients use wearable devices to record their vital signs (heart rate, blood pressure, temperature, etc.), and the data is automatically sent to the hospital's system. If the system detects abnormal values, it generates an alert and forwards it to the doctor for review. Doctors can then provide recommendations or schedule follow-ups with the patients. The provided document below is poorly structured and unclear. Your task is to analyze it and recommend improvements based on software design documentation principles. The RHMS is

intended to monitor patient health metrics remotely (such as heart rate, blood pressure, and temperature) and alert doctors in case of abnormalities.

Instructions:

In this task, you will convert your previous class diagram into actual code using the RHMS scenario. Your goal is to make the design come alive through clean, object-oriented code. If you will be using java programming language, for example:

1. **Set Up Your Project:** Create a new Java project named 'RemoteHealthMonitoringSystem' and separate .java files for each class.
2. **Implement Your Classes:** Define attributes and methods as shown in your UML diagram.
3. **Show Relationships:** Demonstrate inheritance and associations between classes.
4. **Demonstrate Interactions:** In your Main class, show how objects interact to simulate system behavior.
5. **Document Your Work:** Add brief comments explaining class roles and include a short reflection (half a page) on what was easy or difficult to implement.
6. **Upload your codes to Github and add the link in the file.**

Remote Health Monitoring System

The Remote Health Monitoring System is designed to enable doctor's provide medical recommendation to diabetic patient remotely, the patient wears an IoT device which automatically take the patient's readings such as hearRate, temperature, and bloodPressure and send this information to the doctor when abnormal reading is detected.

The aim of this system is to eliminate the time spent going to hospital and provide autonomy for chronic patients.

The patient class, this include every information regarding the patient such name, age, illness, gender, and contactinfo.

The doctor class, this include every information regarding the doctor such as name, are of specialization, and contact information.

The Health information class contain information such as heart rate, temperature, and blood pressure.

The doctor's methods review the data, send feedback and check general health.

Generally it was not so difficult writing the code for the remote health monitoring system, however, i found it little challenging when to comes to the showing of relationship between the classes.